Electric Vehicle Scheduling Model with Strategic Siting and Incentive Pricing Considering Participation Willingness

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***Abstract*—This paper investigates the dispatch process of electric vehicles (EVs) when virtual power plants (VPPs) respond to electricity market demand and proposes an EV dispatch model that considers willingness to participate. The model considers the range anxiety and battery aging anxiety of EVs in the response scheduling process, which effectively improves the willingness of EVs to respond to scheduling. The model mainly involves three aspects: charging station siting, determination of EV incentive electricity price and realization of distributed dispatching. First, this paper proposes a three-step weighted mean clustering method to plan charging station siting, which reduces the driving distance of EVs when responding to grid dispatching, thus reducing range anxiety. At the same time, we studied the effect of different modifications in practical applications. Subsequently, we established an incentive price model for EVs to achieve reasonable quantitative compensation for the battery aging problem that occurs during the dispatch process. Finally, we propose a distributed scheduling strategy to improve the willingness of EVs to respond. This study is validated using real-world data from the city of Chicago. The proposed method reduces the average response distance of EVs in Chicago by 0.188 km and increases the participation willingness of EVs by 0.3243. These results demonstrate the advantages of the proposed approach in alleviating range anxiety and enhancing EVs' willingness to participate in grid dispatch, offering valuable insights for large-scale EV integration into grid interactions.**

***Index Terms*—Electric vehicle, Scheduling, Participation willingness, Charging stations planning, Battery aging**

NOMENCLATURE

Variables

$\boldsymbol{x}_i$	The i-th sample point
$\boldsymbol{\mu}_j$	Center of the j-th cluster
d_{Manh}	Manhattan distance
$d_i^{\max}$	Maximum distance within the partition
d_{ij}	Distance between i-th sample and j-th clustering center
c_i	Cluster center to which the i-th sample belongs
n_j	Number of samples in the j-th cluster
T_{max}	Maximum number of iterations
K	Number of clusters
k_i	Number of initial centers
K'	New number of clusters
S_i	Average distance from the samples in the i-th cluster to their center
β_i	Weight of the i-th samples
w_j	Weight of each cluster center
N_{ex}	Expected number of charging stations
M_{total}	New dataset containing all cluster centers
$\boldsymbol{v}_k$	New cluster center
C_k	Number of cluster centers contained within new k-th cluster
De^{pri}	Priority parameter
De^{N}	Rank order of new charging stations based on number of EVs near charging stations
De^{S}	Rank order of new charging stations based on adjustment distance
S_i^{Change}	The i-th adjusted distance
M_t	Incentive electricity price of EV at time t
ΔM	Adjusted value of dispatch prices
M_t^0	Conventional dispatch price
M_c^0	Basic charging price
$Loss_t$	battery aging compensation price
DOD_t	Discharge depth of the battery at time t
E_t^{start}	The energy of the battery at the beginning of discharge
E_t^{end}	The energy of the battery at the end of discharge
E^m	The rated capacity of the battery
N_{DOD100}^{Total}	Number of times that the battery can discharge when DOD=100%
N_{DOD100}^{eq}	Equivalent cycle number
$N_{DOD100}^{battery}$	The total number of times the battery can be charged and discharged
N^{EAll}	The number of EVs notified for scheduling
N^{ENeed}	The actual amount of EVs dispatched
N^{All}	Total amount of charging stations dispatched
$Loss^{per}$	Equivalent life loss of each discharge
$M^{battery}$	The price of the battery
l_{HS}	Number of high SOC
l_{MS}	Number of medium SOC
l_{LS}	Number of low SOC
D_i^t	Distance between the i-th EV and the charging station
RcI_n^t	The evaluation index
P_n^e	Total allowable charging or discharging capacity of the charging station
P_n^{occ}	Capacity of EVs in the n-th charging station for regular charging
P_n^{free}	Capacity available at n-th charging stations to cover regular charging needs
P^{res}	Response power
P^{dis}	Allocated capacity
P^{Dem}	The total electricity demand dispatched by VPP
P_i	Power consumption per kilometre of the i-th EV
SOC_i^t	SOC of the i-th EV at time t
$Dp_i^{n^{Dp}}$	Distance travelled by the ith EV on the n^{Dp}-th passenger-carrying trip before time t of the day
$De_i^{n^{De}}$	Distance travelled by the i-th EV on the n^{De}-th empty vehicle before the time t of the day
Np_i^t	Number of times the i-th EV carried passengers before the time t of the day
Ne_i^t	Number of times the i-th EV runs empty before the time t of the day
T_1	Number of iterations in the first clustering steps
T_2	Number of iterations in the second clustering steps
T_3	Number of iterations in the third clustering steps

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GS	The grid status
Sur	Surplus power generation
Sho	Shortage power generation
Equ	Grid power balance
H	High SOC EV
L	Low SOC EV
M	Medium SOC EV
μ_k	Sample mean of the k-th column
σ_k	Standard deviation of the k-th column
Parameters	
ε_1	First step clustering preset threshold
ε_2	Second step clustering preset threshold
ε^0	The dispatch margin
η^{DAl}	The distance coefficient
η^{Sta}	The coefficient of charging stations
kp	Equivalent number of cycles
λ	Prioritization coefficient
δ_{jk}	Indicator function
κ	Price adjustment factor
α	*Distance weighting factor*
β	*Price weighting factor*
$\beta_{high}^{surplus}$	Coefficient of EV with high SOC during the period of surplus power generation
$\beta_{medium}^{surplus}$	Coefficient of EV with medium SOC during the period of surplus power generation
$\beta_{low}^{surplus}$	Coefficient of EV with low SOC during the period of surplus power generation
$\beta_{high}^{shortage}$	Coefficient of EV with high SOC during the period of shortage power generation
$\beta_{medium}^{shortage}$	Coefficient of EV with medium SOC during the period of shortage power generation
$\beta_{low}^{shortage}$	Coefficient of EV with low SOC during the period of shortage power generation
$\beta_{high}^{equilibrium}$	Coefficient of EV with high SOC during the period of grid power equilibrium
$\beta_{medium}^{equilibrium}$	Coefficient of EV with medium SOC during the period of grid power equilibrium
$\beta_{low}^{equilibrium}$	Coefficient of EV with low SOC during the period of grid power equilibrium
$\beta_{l_{HS}}$	Dispatch weight coefficients of high SOC EV
$\beta_{l_{MS}}$	Dispatch weight coefficients of medium SOC EV
$\beta_{l_{LS}}$	Dispatch weight coefficients of low SOC EV

I. INTRODUCTION

Electric vehicles (EVs) have significant advantages in alleviating global energy shortages and environmental degradation [1]. With the development of Vehicle-to-Grid (V2G) technology, there is a clear trend of using EVs to exchange electricity with the grid [2], which has become a typical distributed resource. However, the rapid growth of EVs has a significant impact on the planning and operation of the grid [3]. The addition of a large number of EVs will exacerbate the peak-to-valley difference in grid demand, and the uncertainty of EVs will bring challenges to the optimal control of the grid [4]. Virtual power plant (VPP) technology can effectively solve the scheduling and control challenges caused by the differences in geographic location, operating characteristics, and response speed of different types of distributed resources [5], [6]. The benefit of EV participation in VPP is that it can reduce the fluctuation of clean energy sources such as wind power and photovoltaic power [7]. Also, EVs have characteristics that differ from other types of distributed resources, such as mobility, dual identity of production and consumption, and low investment costs [8]. Therefore, it is necessary to study the EV dispatching strategy under the VPP model and consider the characteristics of EVs.

Current research on the participation of EVs in grid scheduling mainly aims at grid stability and economy. In [9], a centralized scheduling method with the goal of reducing EV charging costs is proposed. In [10] and [11], a hierarchical scheduling model for EVs is proposed by considering shared charging stations and demand-side management techniques, respectively. [12] proposes a novel dynamic navigation and charging strategy that fully considers coupled transportation and distribution networks. The above studies effectively reduced the voltage fluctuation [9] and load fluctuation [10]-[12] during EV charging, respectively. However, the above studies lack the overall consideration of EV charging and discharging in response to grid scheduling. In [13], a scheduling mechanism based on an improved charging and discharging opportunity method is proposed to reduce the total charging cost effectively. [14] proposes an urban Internet EV parking system for V2G scheduling, which solves the problem of Internet allocation and scheduling of EVs. In [15], a risk-averse dynamic V2G decision-making algorithm is utilized to effectively improve the expected profit of the parking lot operator. The above study considered the EV discharge process to the grid in the dispatching of EVs. However, the current willingness of EVs to participate in grid scheduling is low [16]. The willingness of EVs to participate in scheduling should also be fully considered in the EV scheduling process, and the current low willingness of EVs to respond to the grid is mainly affected by the aspects of range anxiety and battery aging anxiety [16], [17].

EVs typically need to exchange electrical energy through power exchange points (such as charging stations, home charging facilities, etc.) during response scheduling. Being too far from these power exchange points can increase range anxiety, making their location crucial. Planning the location of power exchange points for EVs with the goal of improving economy and charging reliability is currently a major research focus. In [18], a charging station service area division method based on the Voronoi diagram is first proposed, which effectively improves the economy of charging station planning. [19] improved the economics of EV charging station planning by using charging station cost and user economic loss as planning objectives. Some studies have also combined the grid with the transportation network to consider traffic congestion in EV siting planning. In [20], a robust optimization model for siting distributed energy charging stations based on the combination of road network and power grid is proposed which effectively improves the economics of siting. [21] consideres driver demand and road traffic speed in the development of EV charging scheduling strategies to solve the problem of localized traffic congestion and improve the security and economy of the power grid. In [22], the charging demand points are analyzed and a clustering algorithm is used to realize the planning of EV charging stations. The above studies effectively improve the convenience and economy of EV charging. However, based on the significance of EVs participating in grid scheduling[23], [24], the implementation of national policy documents[25], [26],

[27], and the application of demonstration projects[28], [29], the demand for EVs to discharge to the grid will gradually increase. Current studies lack consideration of this. Comprehensively reducing the distance loss of EVs participating in grid dispatching can effectively improve the willingness of EVs to participate in dispatching.

The purpose of EV users responding to grid dispatch is to gain profit, so setting a reasonable dispatch price is very important for dispatching EVs [30], [31]. Current research on electricity price setting is mainly aimed at improving the interests of operators. [32] and [33] propose methods for charging service pricing and energy management. In [33], a real-time pricing method is proposed and an interval optimization approach is introduced to cope with market price uncertainty. [34] models regional tariffs using a multi-objective optimization algorithm to ensure operators' interests. In [35], a stochastic programming framework is proposed for determining the electricity sales price of an electricity supplier. However, the willingness of EVs to participate in scheduling is related to the profit of EV users, so the profit of EV users also needs to be studied. Currently, EV battery aging anxiety is the main reason for the low willingness of EVs to participate in dispatch. Therefore, battery aging compensation should be considered in the electricity price setting to increase the willingness of EVs to participate in dispatching. In [36], a rule-based V2G scheme is investigated to facilitate charging management in renewable energy systems. However, their strategies did not fully consider battery aging and mileage anxiety due to system complexity constraints. [37] proposes a co-optimization method for economic returns and allocated power that takes into account EV battery degradation models when optimizing EV aggregator revenues. Although the method considers battery depletion in the pricing process, the optimization objective is to maximize the benefits of EV aggregators. In order to improve the willingness of EV users to participate in dispatching, the study of battery loss compensation for EV users should also be carried out, and at the same time, reducing the complexity of the model will help the users to recognize the compensated electricity price, which can improve their willingness to respond [16].

Enhancing the willingness of EVs to participate can increase the likelihood of their response under lower dispatch prices. This, in turn, enables the achievement of the same dispatch objectives at a lower incentive cost, thereby improving the economic efficiency of the dispatch strategy. In this paper, the willingness of EVs to participate in scheduling considering multiple factors is added to the scheduling model for the first time. The willingness of EVs to participate in scheduling is improved by mitigating the negative effects of factors such as range anxiety and battery aging anxiety during EV response. The effectiveness of the proposed method is validated using actual vehicle travel data [38] and charging stations[39] in Chicago, USA. Compared with existing studies, the innovations and contributions of this study can be summarized as:

- A three-step weighted K-means clustering algorithm is proposed for charging station site planning. Compared with existing charging station planning strategies, the algorithm comprehensively reduces the driving distance of EVs responding to scheduling under various types of scheduling scenes by taking into account the relationship between scheduling demand and EV state of charge (SOC) at different moments, which reduces the mileage anxiety of EVs in the response process.
- A distributed scheduling strategy centered on charging stations is proposed, which realizes a scheduling strategy that takes into account multiple factors such as EV location by analyzing the EV state at the scheduling moment. Due to the stochastic nature of EV's response. Compared with the existing scheduling methods, this method can effectively reduce the problem of low response fulfillment rate caused by repeated responses of EVs in the response process. Thus, the response willingness of EVs is improved.

The rest of the paper is organized as follows: Section II presents an overview of the EV scheduling model with strategic site selection and incentive pricing considering participation willingness, Section III presents a planning method for charging stations based on a 3-step weighted k-means method, Section IV proposes an EV incentive price model and a distributed dispatch strategy considering battery loss, Section V uses Matlab to carry out experimental verification and. The paper concludes in Section VI.

II. Overview of The EV Scheduling Model

This paper studies the scheduling of EVs by VPPs in the process of VPP response grid scheduling. Aiming at the problem of low willingness of EVs to respond to scheduling, EV users are unwilling to participate in scheduling for charging and discharging operations based on range anxiety and battery aging anxiety. In this paper, we propose a scheduling model that considers the response willingness of EVs as shown in Fig.1. The scheduling model mainly includes three parts: charging station site selection planning, scheduling electricity price setting and scheduling strategy. As there is a link between the location of charging stations and the scheduling of EVs. The efficiency of EV scheduling is highly dependent on the location of charging stations. If the charging infrastructure is not properly planned, it may increase vehicle range anxiety and battery aging, which may reduce the willingness of vehicle owners to participate in scheduling. Therefore, this paper investigates these two issues together to ensure that the charging infrastructure layout can effectively support EV scheduling.

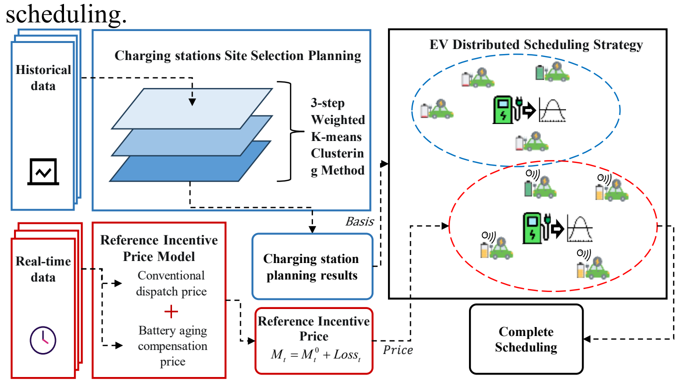


Fig. 1. Overview of the EV scheduling model

EV range anxiety is one of the main reasons for the low willingness of EVs to respond to scheduling, so this paper designs a charging station siting method to reduce range anxiety by reducing the distance that EVs need to consume to respond to grid scheduling. Since grid scheduling includes two modes of discharging and charging, the siting method proposed in this paper takes into account the relationship between the SOC state of EVs and the grid demand. When the grid needs EVs for discharging, the cars with high and medium SOC are made close to the charging station to increase the willingness of high and medium SOC cars to respond to grid scheduling. When the grid needs EV charging, make the low and medium SOC cars close to the charging station to increase the willingness of the low and medium SOC cars to respond to the grid dispatch.

Battery aging caused by charging and discharging operations is another primary reason that affects EV participation in the dispatching process when EVs respond to grid scheduling. Therefore, this paper studies the incentive electricity price for EV dispatching and proposes an incentive electricity price model that considers battery aging. The model is mainly based on the battery life information and the power information of a single dispatch, which reduces the battery aging anxiety of EV users by reasonably compensating the battery aging during the dispatch process and thus improves the willingness of EVs to participate.

Finally, due to the strong decentralized nature of EVs, and at the same time, whether to accept the dispatch is influenced by the subjective judgment of EV users. Therefore, the VPP needs to set a certain dispatching margin in the scheduling process. Too large a scheduling margin will also reduce the realization rate of EV response, thus reducing the willingness of EVs to respond. Therefore, this paper proposes a distributed scheduling strategy, which improves the EV response realization rate while ensuring the scheduling demand, thus increasing the willingness of EVs to participate in scheduling.

III. CHARGING STATIONS SITE SELECTION PLANNING

First, according to the state of power grid operation, we divide 24 hours a day into three situations: grid power surplus, equilibrium, and shortage. Power surplus indicates that the power generation exceeds the load; power shortage indicates that the power generation is less than the load; and power equilibrium represents a state where generation and load are balanced. At the same time, we divide EVs into three categories: high SOC, medium SOC, and low SOC.

Range anxiety is a major issue in the use of EVs, and as a result, many current EV charging station planning strategies aim to minimize the distance during charging[40]. Table I presents the optimization objectives for the distances between various types of EVs and charging stations under different grid conditions. Conventional planning refers to the commonly used strategy that only considers the convenience of charging for EVs, minimizing the distance between EVs and charging stations, thus determining the layout of charging stations. Therefore, regardless of the grid state, the optimization goal is to minimize the distance between EVs needing charging and the nearest charging station[41]. VPP planning refers to a planning strategy that considers the scenario where EVs aggregate as a VPP and participate in grid scheduling. In this scenario, EVs not only charge but also discharge to the grid, depending on the grid's operating status. When power generation exceeds the load, the grid schedules medium and low SOC EVs to charge, increasing the load to balance the grid. When power generation is less than the load, the grid schedules high and medium SOC EVs to discharge, increasing generation to balance supply and demand. Therefore, the optimization goal becomes minimizing the distance between EVs needing to respond to grid scheduling and the nearest charging station. Since EVs with low SOC need to charge, charging during a power shortage would exacerbate the grid imbalance, so the planning needs to maximize the distance between low SOC EVs and the nearest charging station during power shortages, thereby delaying their charging behavior. This strategy falls under flexible guidance rather than mandatory restriction. EVs with low SOC can still opt for nearby charging but will need to bear the additional travel distance. Therefore, their right to charge is not deprived. Instead, by increasing the cost associated with charging during peak load periods, the strategy aims to reduce the willingness of EVs to charge at those times, thereby lowering the overall probability of charging.

TABLE I
OPTIMIZATION TARGET OF THE DISTANCE BETWEEN EVS AND CHARGING STATIONS UNDER DIFFERENT GRID CONDITIONS

Grid state	EV state	Distance	
	SOC	Conventional planning[40], [41]	VPP planning
Power Surplus (power generation exceeds the load)	**High**	-	-
	Medium	-	Min
	Low	Min	Min
Power Shortage (power generation is less than the load)	**High**	-	Min
	Medium	-	Min
	Low	Min	Max
Power Equilibrium (power generation and load are balanced)	**High**	-	Min
	Medium	-	Min
	Low	Min	Min

It can be seen from Table I that different from conventional charging stations planning, the SOC information of EVs needs to be considered in the charging station planning under the VPP scenario. Therefore, we use EV location data at different moments to plan charging stations.

A. 3-step Weighted K-means Clustering Method

Considering that the EV sample size is too large and different SOCs have different influences on site selection, this paper proposes a 3-step weighted k-means clustering method. The structure of the method is shown in Fig. 2.

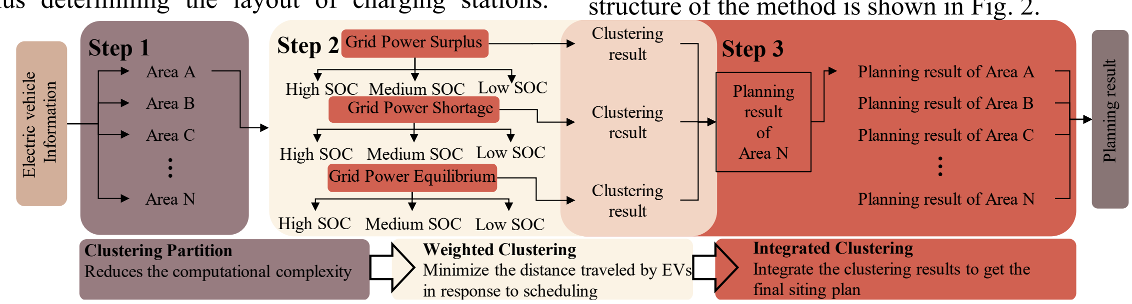


Fig. 2. 3-step weighted k-means clustering method

The 3-step weighted k-means clustering method mainly consists of three steps:

1) Step 1: Clustering Partition

First, we cluster EV location data across all time periods to achieve regional partitioning, effectively reducing data volume and computational load. Given the unique nature of vehicle travel routes, this method utilizes Manhattan distance for calculations. For any two points, $\boldsymbol{x}_i = (x_i, y_i)$ and $\boldsymbol{x}_j = (x_j, y_j)$, the Manhattan distance is defined as:

$$d_{Manh}(\boldsymbol{x}_i, \boldsymbol{x}_j) = |x_i - x_j| + |y_i - y_j| \tag{1}$$

We randomly select K sample points as the initial cluster centers. For each sample point $\boldsymbol{x}_i$, we calculate its Manhattan distance to all cluster centers:

$$d_{ij} = \left\| \boldsymbol{x}_i - \boldsymbol{\mu}_j^{(t-1)} \right\|_1 = \left| x_i - \mu_{j,x}^{(t-1)} \right| + \left| y_i - \mu_{j,y}^{(t-1)} \right| \tag{2}$$

Assign each sample point to the category of the nearest cluster center:

$$c_i = \arg\min_j d_{ij} \tag{3}$$

Update each cluster center to be the mean of the points assigned to that cluster:

$$\boldsymbol{\mu}_j^{(t)} = \frac{1}{n_j} \sum\nolimits_{\boldsymbol{x}_j \in C_i} \boldsymbol{x}_j \tag{4}$$

where n_j is the number of samples in the j-th cluster. The steps of sample assignment and updating cluster centers are repeated until the change in cluster centers is less than a preset threshold ε_1 or the maximum number of iterations T_{max} is reached. The termination condition is as follows:

$$\max_j \left\| \boldsymbol{\mu}_j^{(t)} - \boldsymbol{\mu}_j^{(t-1)} \right\| < \varepsilon_1 \tag{5}$$

$$\left\| \boldsymbol{\mu}_j^{(t)} - \boldsymbol{\mu}_j^{(t-1)} \right\| = \sqrt{(\mu_{j,x}^{(t)} - \mu_{j,x}^{(t-1)}) - (\mu_{j,y}^{(t)} - \mu_{j,y}^{(t-1)})} \tag{6}$$

$\mu_{j,x}^{(t)}$ and $\mu_{j,y}^{(t)}$ represent the longitude and latitude of $\boldsymbol{\mu}_j^{(t)}$, respectively. To determine the appropriate number of clusters K, we use the Davies-Bouldin Index (DBI) to evaluate clustering performance across different values of K.

$$DBI = \frac{1}{K} \sum\nolimits_{i=1}^{K} \max_{j \neq i} \left[\left(S_i + S_j \right) / \left\| \boldsymbol{\mu}_i + \boldsymbol{\mu}_j \right\| \right] \tag{7}$$

where S_i is the average distance from the samples in the i-th cluster to their center:

$$S_i = \frac{1}{n_i} \sum\nolimits_{\boldsymbol{x}_j \in C_i} \left\| \boldsymbol{x}_j - \boldsymbol{\mu}_i \right\|_1 \tag{8}$$

n_i is the number of samples in the i-th cluster, $\boldsymbol{x}_j \in C_i$ indicates that sample $\boldsymbol{x}_j$ belongs to the i-th cluster C_i, and $\boldsymbol{\mu}_i$ is the center of the i-th cluster.

$$\boldsymbol{\mu}_i = \frac{1}{n_i} \sum\nolimits_{\boldsymbol{x}_j \in C_i} \boldsymbol{x}_j \tag{9}$$

$\|\boldsymbol{x}_j - \boldsymbol{\mu}_i\|_1$ represents the Manhattan distance between the sample point and the cluster center. By iterating through different values of K and calculating the corresponding DBI, we select the K that minimizes the DBI as the optimal number of clusters. Through this clustering process, the entire area is divided into K sub-regions.

2) Step 2: Weighted Clustering

In the second step, we apply weighting to each EV within each partition by considering the EV's SOC and the grid status, thereby enhancing the traditional K-means clustering method. This approach allows for a more accurate reflection of the influence of different EVs on charging station site selection.

For each partition, we first initialize the cluster centers by selecting k_i initial centers $\{\boldsymbol{\mu}_1^{(0)}, \boldsymbol{\mu}_2^{(0)}, \ldots, \boldsymbol{\mu}_{k_i}^{(0)}\}$, where k_i is determined based on the proportion of the maximum distance within the partition. First, we calculate the maximum distance within the partition:

$$d_i^{\max} = \max_{\boldsymbol{x}_a, \boldsymbol{x}_b \in C_i} \left\| \boldsymbol{x}_a - \boldsymbol{x}_b \right\|_1 \tag{10}$$

Then, we determine the number of clusters:

$$k_i = \left\lceil \left(d_i^{\max} / \sum\nolimits_{j=1}^{K} d_j^{\max} \right) \times N_{ex} \right\rceil \tag{11}$$

where N_{ex} is the expected number of charging stations. For each sample point $\boldsymbol{x}_i$, we calculate its weighted Manhattan distance to all cluster centers:

$$d_{ij} = \beta_i \times \left\| \boldsymbol{x}_i - \boldsymbol{\mu}_j^{(t-1)} \right\|_1 \tag{12}$$

β_i represents the weight, and the weight β_i for each sample point $\boldsymbol{x}_i$ is determined by a combination of the EV's SOC and the grid status:

$$\beta_i = \begin{cases} \beta_{high}^{surplus} & (GS = Sur, SOC_i = H) \\ \beta_{medium}^{surplus} & (GS = Sur, SOC_i = M) \\ \beta_{low}^{surplus} & (GS = Sur, SOC_i = L) \\ \beta_{high}^{shortage} & (GS = Sho, SOC_i = H) \\ \beta_{medium}^{shortage} & (GS = Sho, SOC_i = M) \\ \beta_{low}^{shortage} & (GS = Sho, SOC_i = L) \\ \beta_{high}^{equilibrium} & (GS = Equ, SOC_i = H) \\ \beta_{medium}^{equilibrium} & (GS = Equ, SOC_i = M) \\ \beta_{low}^{equilibrium} & (GS = Equ, SOC_i = L) \end{cases} \tag{13}$$

$$s.t. \begin{cases} 0 = \beta_{high}^{surplus} < \beta_{medium}^{surplus} \leq \beta_{low}^{surplus} \\ \beta_{low}^{shortage} < 0 < \beta_{medium}^{shortage} \leq \beta_{high}^{shortage} \\ 0 < \beta_{high}^{equilibrium} = \beta_{low}^{equilibrium} \leq \beta_{medium}^{equilibrium} \end{cases} \tag{14}$$

GS represents the grid status, which includes three conditions: Sur indicates grid power surplus, Sho indicates grid power shortage, and Equ indicates grid power balance. SOC_n denotes SOC of the n-th EV, comprising three levels: H for high SOC, L for low SOC, and M for medium SOC.

$$\mathrm{GS} \in \{Sur, Sho, Equ\} \tag{15}$$

$$\mathrm{SOC}_i \in \{H, L, M\} \tag{16}$$

When the clustering center is required to be close to a specific EV, the weight of that EV is set to 1. When the clustering center is expected to be distant from certain EVs, their weights are set to -1. If an EV is not considered in the clustering center determination, its weight is set to 0.

When the power grid requires EV discharging, both high-SOC and medium-SOC EVs can participate in the dispatch, so their weights are set to 1. Low-SOC EVs not only cannot contribute to discharging, but their charging would further aggravate grid imbalance; therefore, their weight is set to -1. When the grid requires EV charging, medium-SOC and low-SOC EVs can participate, and their weights are set to 1, while high-SOC EVs cannot participate and are thus assigned a weight of 0. In a power equilibrium scenario, there is no need for EV dispatch, and all EVs are assigned the same weight of 1.

For a unified formulation, the weights are expressed in Eq. (17). In this equation, the first deduction term takes the value of 2 for low-SOC EVs during a power shortage, reducing their weight to -1, thereby excluding vehicles unfavorable for discharging. The second deduction term takes the value of 1 for high-SOC EVs during a power surplus, making their weight 0, meaning they neither attract nor repel the clustering center.

$$\beta_i = 1 - 2 \cdot \mathbf{1}[GS = Sho] \cdot \mathbf{1}[SOC_i = L] - \mathbf{1}[GS = Sur] \cdot \mathbf{1}[SOC_i = H] \tag{17}$$

where $\mathbf{1}[\cdot]$ denotes the indicator function, which takes the value 1 if the condition inside the brackets is satisfied, and 0 otherwise. The SOC is categorized according to the following formula:

$$\begin{cases} low\ SOC\ EV & (0 \le SOC < 40\%) \\ Medium\ SOC\ EV & (40\% \le SOC \le 60\%) \\ high\ SOC\ EV & (60\% < SOC \le 100\%) \end{cases} \tag{18}$$

Assign each sample point to the category of the cluster center with the minimum weighted distance:

$$c_i = \arg\min_j d_{ij} \tag{19}$$

Then, update each cluster center to be the weighted average of the points assigned to that cluster:

$$\boldsymbol{\mu}_j^{(t)} = \sum\nolimits_{\boldsymbol{x}_j \in C_j} \beta_i \times \boldsymbol{x}_j \Big/ \sum\nolimits_{\boldsymbol{x}_j \in C_j} \beta_i \tag{20}$$

Repeat the steps of weighted sample assignment and updating cluster centers until the change in cluster centers is less than the preset threshold ε_2 or the maximum number of iterations is reached. The termination condition is as follows:

$$\max_j \left\| \boldsymbol{\mu}_j^{(t)} - \boldsymbol{\mu}_j^{(t-1)} \right\| < \varepsilon_2 \tag{21}$$

3) Step 3: Integrated Clustering

The third step is to integrate the results of the three layers of grid power surplus, shortage, and equilibrium. In this step, we use the number of original samples to assign values to the weights and then re-cluster the clustering centers of each layer to get the final clustering results.

For each cluster center $\boldsymbol{\mu}_j$, assign a weight w_j , which is equal to the total weight of the sample points corresponding to that cluster center:

$$w_j = \sum\nolimits_{x_i \in C_j} \beta_i \tag{22}$$

Combine all the partitioned and layered cluster centers to form a new dataset:

$$M_{total} = \bigcup\nolimits_{i=1}^{K} \bigcup\nolimits_{l=1}^{L} \{\boldsymbol{\mu}_j^{(i,l)}\} \tag{23}$$

where $\boldsymbol{\mu}_j^{(i,l)}$ represents the j-th cluster center in partition i and layer l. Perform weighted k-means clustering on the merged set of cluster centers M_{total} to further reduce the number of cluster centers. The objective function is:

$$J_{final} = \sum\nolimits_{k=1}^{K'} \sum\nolimits_{\boldsymbol{\mu}_j \in M_{total}} w_j \times \delta_{jk} \times \left\| \boldsymbol{\mu}_j - \boldsymbol{v}_k \right\|^2 \tag{24}$$

$\boldsymbol{v}_k$ represents the new cluster center, and K' is the new number of clusters, usually less than $|M_{total}|$. δ_{jk} is an indicator function, where δ_{jk}=1 if cluster center $\boldsymbol{\mu}_j$ is assigned to the new cluster center $\boldsymbol{v}_k$, otherwise, δ_{jk}=0. Then, update the cluster centers:

$$\boldsymbol{v}_k = \sum\nolimits_{\boldsymbol{\mu}_j \in C_k} w_i \times \boldsymbol{\mu}_j \Big/ \sum\nolimits_{\boldsymbol{\mu}_j \in C_k} w_i \tag{25}$$

where C_k is the set of cluster centers contained within the new k-th cluster. Repeat the steps of re-clustering the weighted cluster centers and updating the final cluster centers until the change in cluster centers is less than the preset threshold or the maximum number of iterations is reached.

To evaluate the computational efficiency of the proposed three-step weighted k-means clustering algorithm, both time and space complexities are analyzed.

In Step 1, the standard k-means algorithm is used for initial spatial partitioning of N EV samples into K clusters, with a time complexity of $O(NKT_1)$. In Step 2, weighted k-means clustering is applied within each partition. Assuming the average number of clusters per partition is $\bar{k}$, the time complexity is $O(N\bar{k}\ T_2)$. In Step 3, all cluster centers from different grid states are merged and clustered again. Let the total number of initial centers be M, and the final number of clusters be K'. As the input size in this step is significantly smaller, the additional time complexity is $O(MK'T_3)$. where T_1, T_2, and T_3 denote the number of iterations in the first, second, and third clustering steps, respectively.

Combining all steps, and considering that M is relatively small, the total time complexity of the algorithm is:

$$O(NKT_1 + N\bar{k}T_2 + MK'T_3) \approx O(N(KT_1 + \bar{k}T_2)) \tag{26}$$

The algorithm mainly stores EV sample data and intermediate cluster centers. With input features of dimension d, the space complexity is:

$$O(N(d + \bar{k})) \tag{27}$$

B. Uncertainty Analysis

In practical scenarios, the spatial distribution, time of appearance, and SOC of EVs are inherently uncertain and correlated. To account for this, a Gaussian Copula-based Monte Carlo simulation method is employed to model the joint uncertainty of these variables and generate synthetic yet realistic EV behavior samples. Assume that the matrix $\mathbf{X}$ is given, which contains the longitude λ, latitude ϕ, charging start time t, and SOC.

$$\mathbf{X} \in \mathbb{R}^{N \times 4} \tag{28}$$

First, Z-score normalization is applied:

$$z_{ik} = \frac{x_{ik} - \mu_k}{\sigma_k} \tag{29}$$

μ_k and σ_k denote the sample mean and standard deviation of the k-th column, respectively. This results in a matrix $\mathbf{Z}$ with zero mean and unit variance. Then, the empirical distribution function is used to transform $\mathbf{Z}$ into pseudo-observations on the unit interval. Subsequently, a 4-dimensional Gaussian copula is employed to describe the dependence structure among the variables.

$$C_\Sigma(\mathbf{u}) = \Phi_\Sigma\left(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_4)\right) \tag{30}$$

where Φ_Σ denotes the cumulative distribution function of a zero-mean multivariate normal distribution. The parameter matrix Σ can be estimated using the maximum likelihood method.

Then, Monte Carlo sampling is used to generate data. First, n samples s_{jk} are drawn from the multivariate normal distribution $\mathcal{N}(0, \Sigma)$. Each element is then transformed using the standard normal cumulative distribution function:

$$u_{jk} = \Phi(s_{jk}) \tag{31}$$

Finally, according to:

$$x_{jk} = u_{jk}\sigma_k + \mu_k \tag{32}$$

the synthetic matrix $\mathbf{X} \in \mathbb{R}^{n\times 4}$ is obtained.

C. Priority Adjustment Method

In practice, since most cities already have a large number of charging stations, the charging station optimization should also incorporate information of existing charging stations to reduce the workload and thus increase the practicality of the method. Therefore, this paper proposes the adjustment of the priority parameter De^{pri}. The adjustment of the priority takes into account the number of EVs near the new charging station and the original charging station situation near the new charging station.

$$De^{Pri} = De^{N} + \lambda De^{S} \tag{33}$$

where De^N is the rank order of new charging stations based on N (number of EVs near charging stations). De^S is the rank order of new charging stations based on S (adjustment distance). λ is the prioritization coefficient, where the larger λ is the more important the adjustment distance is.

$$S_i^{Change} = \left\| \boldsymbol{x}_{Ch,i} - \boldsymbol{\mu}_{Or,j} \right\|_1 = \left| x_{Ch}^i - x_{Or}^j \right| + \left| y_{Ch}^i - y_{Or}^j \right| \tag{34}$$

where S_i^{Change} is the i-th adjusted distance, $\boldsymbol{x}_{Ch,i}$ is the i-th adjusted charging station, $\boldsymbol{x}_{Or,i}$ is the nearest original charging station to $\boldsymbol{x}_{Ch,i}$. The smaller the adjustment priority parameter De^{pri}, the higher the priority.

IV. EV Scheduling Strategy

The VPP plays a pivotal role in coordinating the charging and discharging activities of EVs. Acting not only as an aggregator of EV resources but also as a real-time decision-making center, the VPP dynamically schedules EV charging and discharging based on the current grid conditions. When the grid experiences a power surplus, the VPP schedules EVs with low and medium SOC to charge, thereby absorbing the excess energy. Conversely, during power shortages, the VPP coordinates EVs with high and medium SOCs to discharge, feeding electricity back into the grid to help restore the supply-demand balance. To enable real-time dispatch and response collection, this study employs the public cellular network as the primary communication infrastructure. Cellular networks offer wide urban coverage and low latency, aligning well with the requirements of EV dispatch. This facilitates the timely distribution of dispatch commands to individual electric vehicles, ensuring rapid response and minimizing delays. Moreover, leveraging existing commercial networks eliminates the need for dedicated communication infrastructure, thereby enhancing scalability.

A. Incentive Price Model Considering Battery Aging

EV has the characteristics of being rechargeable and dischargeable, a large amount of resources, etc., and is one of the suitable options for VPP composition. However, when VPP schedules EVs, it needs to charge and discharge EV batteries, and regular charging and discharging will have an impact on battery life. This is one of the main reasons that affect the willingness of EVs to respond to VPP[8]. At present, most VPPs have the same settlement policy for EVs as other energy sources. They only settle according to the price of supplied electric energy. Therefore, it is difficult to attract EVs in actual operation. Therefore, it is necessary to research subsidized prices for battery wear and tear.

Set the incentive electricity price of EV at time t as M_t, including conventional dispatch price M_t^0 and battery aging compensation price $Loss_t$, then:

$$M_t = M_t^0 + Loss_t \tag{35}$$

where M_t^0 includes the electricity cost, operation and maintenance cost, time cost and other information of the charging station.

The life of the battery is related to the depth of discharge and the number of cycles.

$$DOD_t = (E_t^{start} - E_t^{end}) / E^m \times 100\% \tag{36}$$

where DOD_t is the discharge depth of the battery at time t. E_t^{start} and E_t^{end} are the energy of the battery at the beginning of discharge and the energy of the battery at the end of discharge respectively. E^m is the rated capacity of the battery.

If the number of times N_{DOD100}^{Total} that the battery can discharge when DOD=100% is set as the life of the battery, the equivalent life loss of each discharge $Loss^{per}$ can be expressed as:

$$Loss^{per} = N_{DOD100} / N_{DOD100}^{Total} \tag{37}$$

And the equivalent DOD^{100} cycle number N_{DOD100}^{eq} of N_{DODd} cycles under DOD^d is [42]:

$$N_{DOD100}^{eq} = N_{DODd} (DOD^d)^{kp} \tag{38}$$

where kp is a constant from 0.8 to 2.1, a larger kp means that the equivalent DOD^{100} cycles of N_{DODd} cycles at DOD^d are less, because $DOD^d \leqslant 100\%$. In the process of VPP dispatching EVs, it is assumed that only the loss of battery life due to the allocated quota needs to be considered, and the loss of battery life due to other reasons such as the journey to charging stations is not considered. The DOD_t^d caused by scheduling at time t can be expressed as the ratio of allocated capacity $P_{t,n}^{dis}$ to rated battery capacity $E_{t,n}^m$.

Since this article is aimed at the scheduling of EVs by VPP, in this mode EV will not alternate between charging and discharging modes, and the number of cycles will not be greater than 1. Therefore, the equivalent number of cycles N_{DODd}^{eq} in this paper can be equivalent to DOD^d.

Then the battery aging compensation price $Loss_t$ caused by VPP scheduling at time t is:

$$Loss_t = (M^{battery} / N_{DOD100}^{battery}) \times (P_{t,n}^{dis} / E_{t,n}^{M})^{kp+1} \tag{39}$$

where $M^{battery}$ is the price of the battery, and $N_{DOD100}^{battery}$ is the total number of times the battery can be charged and discharged. The final incentive price is:

$$M_t = M_t^0 + (M^{battery} / N_{DOD100}^{battery}) \times (P_{t,n}^{dis} / E_{t,n}^{M})^{kp+1} \tag{40}$$

The incentive pricing mechanism proposed in this study, which accounts for battery degradation, consists of three main steps. First, an empirical formula is used to calculate the equivalent battery life loss caused by a single dispatch, considering the impacts of depth of discharge and number of cycles. Then, based on the battery's total cost and usable cycle life, this degradation is converted into a monetary value. Finally, this value is incorporated into the incentive price to ensure that EV owners receive fair compensation for battery wear, thereby enhancing their willingness to participate.

B. EV Distributed Scheduling Strategy

Response willingness of EV is related to response realization rate RaR:

$$RaR = N^{ENeed} / N^{EAll} \tag{41}$$

where N^{EAll} is the number of EVs notified for scheduling, and N^{ENeed} is the actual amount of EVs dispatched. We use the consumer psychology user response model [43] to simulate the relationship between EV response willingness and response realization rate.

Due to the large number of EVs, direct scheduling of all EV users results in low response realization rates and reduces the willingness of EVs to participate in scheduling. Therefore, it is necessary to study the dispatch order. We conduct a statistical analysis of EV information centered on charging stations, and use the set statistical features to sort charging stations to achieve order dispatch.

Both the distance of EVs from charging stations and the number of EVs with different SOCs will affect the responsiveness of charging stations. When VPP needs EVs to provide electric energy, they need high-SOC and medium-SOC EVs for aggregation. Therefore, the more high-SOC and medium-SOC EVs, the higher the ranking of charging stations. Since EVs with low SOC not only cannot provide energy to VPP but also need to perform charging operations, the more the number of EVs with low SOC, the lower the ranking of charging stations. At the same time, the distance between EVs and charging stations will also affect the responsiveness of charging stations. When the distance between EVs with high SOC and medium SOC is close to charging stations, the charging willingness of EVs is higher due to the lower consumption of the journey to charging stations, the ranking of charging stations rises, and vice versa, the ranking decreases. When the EV with low SOC is closer to the charging stations, its willingness to charge increases, but at this time, VPP requires the EV to output electric energy, so the ranking of charging stations decreases, and vice versa, the ranking rises. Therefore, the number of EVs around charging stations and the weight distance can be used to set the statistical evaluation index of charging stations responsiveness RcI_n^t. When the power grid requires EV discharging:

$$RcI_n^t = \beta_{l_{HS}} l_{HS} + \beta_{l_{MS}} l_{MS} + \beta_{l_{LS}} l_{LS} - \frac{1}{\eta^{DAl}} \left(\sum_{i=1}^{l_{HS}} \frac{D_i^t}{l_{HS}} + \sum_{i=1}^{l_{MS}} \frac{D_i^t}{l_{MS}} - \sum_{i=1}^{l_{LS}} \frac{D_i^t}{l_{LS}} \right) \tag{42}$$

where l_{HS}, l_{MS}, l_{LS} represents the number of high SOC, medium SOC, and low SOC respectively. D_i^t represents the distance between the i-th EV and the charging station. $\beta_{l_{HS}}$, $\beta_{l_{MS}}$, $\beta_{l_{LS}}$ represents the weight coefficients of high SOC, medium SOC and low SOC EVs, respectively. η^{DAl} is the distance coefficient, which indicates the emphasis of the distance and the number of EVs. If it is increased, the influence of the distance on the sorting results can be reduced. Otherwise, the influence of the distance can be increased. When the power grid requires EV charging:

$$RcI_n^t = \beta_{l_{MS}} l_{MS} + \beta_{l_{LS}} l_{LS} - \frac{1}{\eta^{DAl}} \left(\sum_{i=1}^{l_{MS}} \frac{D_i^t}{l_{MS}} + \sum_{i=1}^{l_{LS}} \frac{D_i^t}{l_{LS}} \right) \tag{43}$$

The larger the evaluation index RcI_n^t , the better the responsiveness of the charging stations at this moment. According to this index, the responsiveness of the charging stations is sorted. In practical applications, the SOC data of EV is transmitted to the dispatch center via the communication network in real time, thus supporting the dispatch center's calculation and analysis.

The distributed power P_n of each charging station and the total amount of charging stations dispatched N^{All} are as follows:

$$P_n = \eta^{Sta} P_n^e \quad (0 \le \eta^{Sta} \le 1) \tag{44}$$

$$\min N^{All} \quad s.t. \sum_{n=1}^{N^{All}} P_n \ge P^{Dem} + \varepsilon^0 \tag{45}$$

where P_n^e represents the total allowable charging or discharging capacity of the charging station for EVs at the current moment, which refers to the total amount of power that the charging station is able to flow to the EVs at this moment or the total amount of power that it is able to accept the inflow of the EVs. η^{Sta} is the coefficient of charging stations. It ensures that the charging station can preserve a certain margin to provide charging function for normal EVs in the task of consuming power by the EVs. P^{Dem} is the total electricity demand dispatched by VPP. Communication failures may prevent some electric vehicles from receiving dispatch instructions, and even when instructions are successfully received, certain vehicles may still exhibit non-compliant behavior during actual response. These factors introduce uncertainty into the dispatch outcomes and reduce system stability. To enhance system robustness, this study introduces a dispatch margin ε^0, which reserves a certain buffer in dispatch capacity to accommodate unexpected situations. ε^0 is set to a value proportional to P^{Dem}. The value needs to be taken in conjunction with realistic data, such as historical dispatch default rates, to estimate the possible compensation.

During the dispatch process, unnecessary occupation of charging piles can interfere with the implementation of the dispatch strategy. To prevent unnecessary occupation of dispatch charging station resources, this study proposes a charging guidance strategy based on resource status and dynamic pricing. Considering that fast-charging stations typically require less than one hour to fully charge a conventional EV, a one-hour advance control window is established to prevent EVs from occupying charging spots prematurely, which could interfere with the dispatch process. During both the advance control window and the dispatch period, the proposed strategy is applied to regulate EV charging. In this process, the currently available charging capacity reserved for meeting regular charging demand is denoted as：

$$P_n^{free} = \left(1 - \eta^{Sta}\right) P_n^e - P_n^{occ} \tag{46}$$

where P_n^{occ} represents the total capacity of EVs currently undergoing regular charging at the n-th charging station. When P_n^{free} >0, additional EVs are allowed to charge at the station, but they must pay a dynamically increased charging price:

$$M_n^c = M_n^{c0} + \kappa\left(1-\frac{P_n^{free}}{\left(1-\eta^{Sta}\right)P_n^e}\right) \tag{47}$$

where M_c^0 represents the base charging price, and κ is the price adjustment coefficient. The adjusted portion of the price increases as the remaining charging resources decrease. This mechanism filters EVs through pricing, prioritizing those willing to pay higher fees—typically EVs with stronger charging needs—thereby controlling the resource occupancy rate.

If an EV is unwilling to accept the price increase, or when $P_n^{free} \le 0$ and no additional EVs are allowed to access the station, the system redirects the EV to another charging station not involved in the current grid dispatch. The redirection follows a "distance–price" weighted cost minimization criterion.

$$j^* = \arg\min_{j\in S}\left(\alpha\cdot d_{i,j} + \beta\cdot M_j^c\right) \tag{48}$$

where, $d_{i,j}$ represents the distance between i-th EV and candidate charging station j, M_j^c is the charging price at station j, and α and β are the weighting factors for distance and price, respectively. Therefore, the charging dispatch decision for each EV can be uniformly expressed as:

$$Decision_i = \begin{cases} \text{Charge at station } k, & \text{if } P_k^{free}>0 \text{ and } M_k^c \text{ is accepted} \\ \text{Redirect to } j^*, & \text{if } P_k^{free}\le 0 \text{ or } M_k^c \text{ is rejected} \end{cases} \tag{49}$$

This mechanism minimizes unnecessary occupation of dispatch charging piles by restricting access to resources, implementing dynamic price regulation, and selecting optimal guidance paths, thereby ensuring the smooth execution of dispatch demands.

In practical dispatch operations, whether EV users are willing to participate in the response depends not only on the current electricity price but also on heterogeneous factors such as individual behavioral preferences and travel plans. Therefore, a fixed incentive price may be insufficient to dispatch a large enough group of EVs, potentially affecting the success of the dispatch. To address this, this study introduces a flexible price adjustment mechanism [44] that gradually modifies the dispatch price to ensure the successful completion of dispatch tasks.

The initial dispatch price is M_k The system adjusts the dispatch price based on the EV users' response under this price. If the total response energy P^{res} fails to meet the grid-side dispatch demand $P^{Dem}+\varepsilon^0$, the dispatch price is adjusted accordingly:

$$M_{k+1} = M_k + \Delta M \tag{50}$$

where ΔM is the adjustment increment of the dispatch price. The next dispatch iteration is then executed. This process continues until the following termination condition is met:

$$P^{res} \ge P^{Dem} + \varepsilon^0 \tag{51}$$

This mechanism ensures that the response from EV users is sufficient to complete the dispatch task by gradually adjusting the dispatch price, thereby ensuring the practical applicability of the proposed method.

V. Example and Simulation

This paper uses the taxi data set for experimental verification. The data comes from the 2014-2022 Chicago taxi driving data set released by the US public government. Since the dataset does not contain information about SOC directly, in the simulation part of this paper it is calculated based on the cumulative distance traveled by the EV at different times of the day:

$$SOC_i^t = 1-\frac{P_i}{E_i^m}\left(\sum_{n^{Dp}=1}^{Np_i^t} Dp_i^{n^{Dp}} + \sum_{n^{De}=1}^{Ne_i^t} De_i^{n^{De}}\right) \tag{52}$$

where SOC_i^t represents the SOC of the i-th EV at time t. E_i^m is the rated capacity of the i-th EV. P_i is the power consumption per kilometre of the i-th EV. Np_i^t indicates the number of times the i-th EV carried passengers before the time t of the day. $Dp_i^{n^{Dp}}$ indicates the distance travelled by the ith EV on the n^{Dp}-th passenger-carrying trip before time t of the day. Ne_i^t indicates the number of times the i-th EV runs empty before the time t of the day. $De_i^{n^{De}}$ indicates the distance travelled by the i-th EV on the n^{De}-th empty vehicle before the time t of the day. The overall algorithm implementation flow is shown in Fig.3.

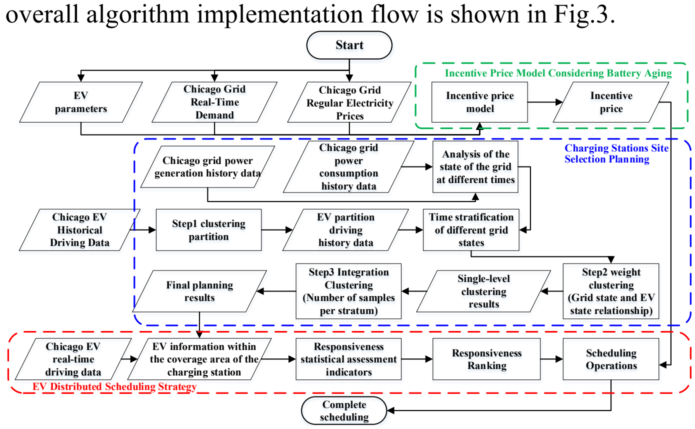
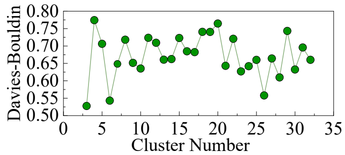


Fig. 3. Overall algorithm implementation flow

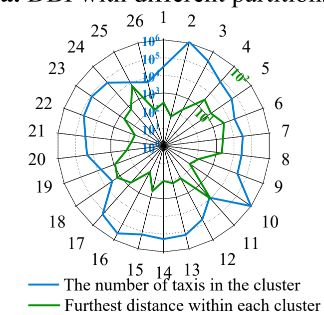
A. *Charging station Planning Simulation*

1) *Sample Partition*

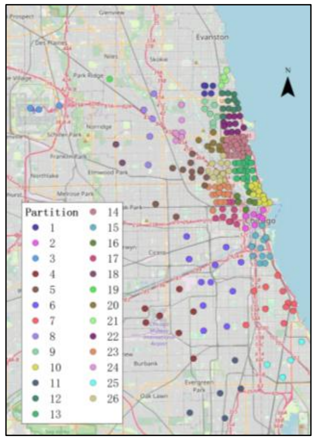
The sample set used in this paper contains about 4 million sample points, and the amount of calculation for direct weight clustering is too large. Therefore, firstly, k-means clustering is used to partition the samples. To determine the appropriate number of partitions, this paper selects 30 partitions, with the number of clusters ranging from 3 to 32 for comparative analysis. The DBI under different number of partitions is shown in Fig.4a. It can be seen that when the number of partitions is 3, the DBI is the smallest, which is 0.5276. However, using 3 as a partition reduces fewer number of subsequent calculations. When the number of partitions is 26, the DBI is 0.5579. Although it is slightly larger than the DBI when the number of partitions is 3, it reduces the number of subsequent calculations significantly. Therefore, the clustering result with 26 partitions is selected to partition the original sample. Fig. 4b is the number of taxis in each partition and the maximum distance between taxis in each partition, and Fig. 4c is the partition result of the Chicago area data.



a. DBI with different partitions



b. Clustering Analysis



c. Clustering results with 26 partitions

Fig. 4. Determination of cluster number and cluster results

2) Sample Stratification

We stratify the partitioned data according to time. PJM is the largest power operator in the United States, and its business scope covers the city of Chicago. Therefore, we use the 2019-2022 power demand and power generation data published by PJM [45] to obtain power demand data and conduct statistical analysis on it. It is found that in the eight sampling time points of 20:00-03:00, there is a probability of more than 70% that the power generation is greater than the power demand (power surplus); in the three sampling points of 06:00-08:00, There is a probability of more than 70% that the power generation is less than the power demand (power shortage). Therefore, 24 hours a day are divided: 20:00-03:00 is the grid power surplus period, 06:00-08:00 is the power shortage period, 04:00-05:00, and 09:00-19:00 is the electric energy equilibrium period.

According to Table I, the weight coefficients are set:

$$\begin{bmatrix} \beta_{high}^{shortage} & \beta_{medium}^{shortage} & \beta_{low}^{shortage} \\ \beta_{high}^{equilibrium} & \beta_{medium}^{equilibrium} & \beta_{low}^{equilibrium} \\ \beta_{high}^{surplus} & \beta_{medium}^{surplus} & \beta_{low}^{surplus} \end{bmatrix} = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad (53)$$

3) Stratification Clustering

The improved weight clustering method is used to cluster the partitioned EV, and the clustering results are shown in Fig. 5. From the clustering results, it can be seen that the improved k-means clustering method can consider the weights so that the sample points with larger weights are closer to the cluster center, and the sample points with smaller weights and samples with negative weights points are farther from the center. Among them, when the power grid is surplus, and medium and low SOC EVs are required for VPP aggregation, the average distances of medium and low SOC EVs from the charging stations in the cluster are 0.0641km and 0.0673km, respectively, much smaller than the average distance of high SOC EVs of 2.413km. When the grid power is short and requires medium- and high-SOC EVs to perform VPP aggregation, the average distances between medium and high SOC EVs and charging stations in the cluster are 0.1070km and 0.0991km, respectively, much smaller than the average distance of low-SOC EVs 3.848 km. It can be seen that the clustering center can be shifted according to different weights by using this method, so that charging stations can be effectively selected according to the relationship between different SOCs and grid states.

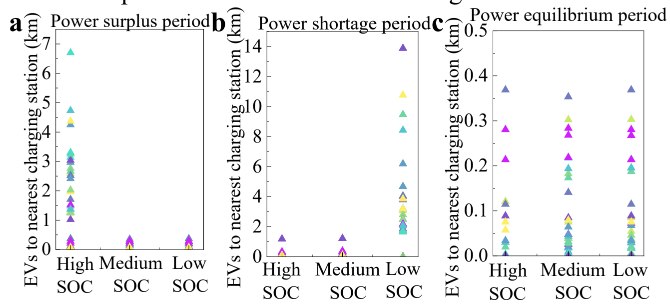


Fig. 5. Clustering results of EV data after stratification

4) Integrated Clustering

After obtaining the stratification clustering results, set the weight according to the number of samples in each layer and cluster the results of the three layers. At the same time, each region is spliced to form the final clustering result, that is, the position of charging stations. To ensure the validity of the subsequent comparative analysis results, this study sets the target number of clustering centers to 292, consistent with the current number of charging stations in the City of Chicago. The results of the comparative analysis using the location information of existing charging stations in Chicago are shown in Fig. 6. Using the method of this paper to plan the location of charging stations, 95% of EVs can reach charging stations within 0.8km, and the average distance of all EV samples to the nearest charging stations is 0.3155km. At present, if the actual charging stations in Chicago achieve 95% EVs coverage, it will take 2km, and the average distance from all EV samples to the nearest charging stations is 0.5035km. Therefore, it can be seen that the method proposed in this paper can effectively reduce the distance of EVs to charging stations and improve the charging and discharging enthusiasm of EVs, especially when the grid electric energy shortage and electric energy surplus require EVs to response.

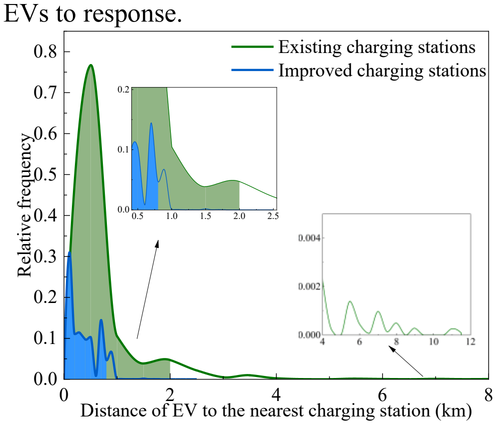


Fig. 6. Final location and analysis of charging stations

Meanwhile, we choose planning based on the cost of charging stations and the economic loss of users in [17], planning based on tochastic dynamic equilibrium in [46] and planning based on flow capture location in [47] for comparison. The results are shown in Table II. Compared with the existing charging stations in Chicago, all of the planning strategies considered significantly reduce the charging distance for EVs. Among them, the method proposed in this paper achieves superior performance in minimizing the response distance of EVs to grid dispatch, with reductions of 1.59%, 24.88%, and

18.39% compared to the other three planning algorithms, respectively. This demonstrates the effectiveness of the proposed method in reducing EV response distances overall, thereby enhancing their willingness to participate in grid dispatch. Although the reduction effect is not obvious compared with [17], EVs with low SOC are harmful to the grid when the grid power is insufficient. Compared with EVs with high SOC when the grid power is insufficient, EVs with low SOC in [17] are closer to the charging station. At this time, EVs with low SOC have a higher willingness to charge, which is not conducive to the stability of the grid. In contrast, under the same grid shortage scenario, the average distance from low-SOC EVs to charging stations in our method is 0.3017 km, which is greater than that of high-SOC EVs. This indicates that low-SOC EVs exhibit lower charging willingness when power is scarce, which is beneficial for grid stability.

TABLE II

COMPARATIVE ANALYSIS OF CHARGING STATIONS LOCATIONS

Period		Power surplus	Power shortage	Power equilibrium	Participate in scheduling	Radius 95
Planning strategy proposed in this paper (km)	High SOC	0.2904	0.2831	0.3275	0.2995	0.7342
	Medium SOC	0.3224	0.2807	0.3377		
	Low SOC	0.3231	0.3017	0.3399		
Existing charging station (km)	High SOC	0.5256	0.5768	0.4867	0.5413	2.1916
	Medium SOC	0.4843	0.6451	0.4731		
	Low SOC	0.5037	0.7909	0.4681		
Planning based on the cost of charging stations and the economic loss of users [48] **(km)**	High SOC	0.2751	0.2983	0.3306	0.3043	0.8643
	Medium SOC	0.3055	0.2813	0.3229		
	Low SOC	0.3066	0.2869	0.3258		
Planning based on tochastic dynamic equilibrium [46] **(km)**	High SOC	0.4051	0.3961	0.4034	0.3987	1.3115
	Medium SOC	0.4026	0.4031	0.402		
	Low SOC	0.4006	0.4144	0.4007		
Planning based on flow capture location [47] **(km)**	High SOC	0.3205	0.3929	0.3371	0.367	1.2856
	Medium SOC	0.3254	0.4316	0.3296		
	Low SOC	0.3402	0.6167	0.3207		

Nevertheless, compared to other methods, the driving distances of low-SOC EVs remain relatively short. This is because the proposed method performs global distance optimization for all vehicles while introducing differentiated guidance across time periods and SOC levels. Specifically, it applies less aggressive distance optimization for low-SOC EVs during power shortage periods, making them relatively farther from charging stations than high- or medium-SOC EVs. But this does not mean that it goes further than the original plan or other methods, it just means that the optimization is smaller compared with the same period. This strategy not only enhances overall EV responsiveness to dispatch but also provides differentiated guidance to various types of EVs across different time slots, thereby improving grid stability.

In addition, we also conducted a 95% coverage analysis. Under the proposed method, the response distance of 95% of the participating electric vehicles was within 0.73 km, which is the smallest among all methods, further verifying the advantage of the proposed method in shortening the response distance.

Since information such as the position of the EV and the SOC state is uncertain, this paper utilizes the Coupled Monte Carlo method [49] to analyze the uncertainty. This method can fully take into account the effects between different characteristics in the sample generation process. 10,000 EV samples containing location, time and SOC state information are generated, and the distances between the EVs in the samples and the nearest charging stations in different scenes are calculated separately. The results are shown in Table III. The average distance from all sample EVs to the nearest charging station is 0.3262km (95% CI:0.3165-0.3359km). It can be seen that the uncertainty of EVs has a small impact on the effectiveness of the siting strategy proposed in this paper, which verifies the reliability of the method proposed in this paper.

TABLE III

DISTANCE FROM EV TO THE NEAREST CHARGING STATION

Scenario	Mean (km)	95%CI Lower (km)	95%CI Upper (km)
High SOC in Power surplus	0.3545	0.3403	0.3688
Medium SOC in Power surplus	0.3182	0.3041	0.3323
Low SOC in Power surplus	0.3320	0.3144	0.3497
High SOC in Power shortage	0.2906	0.2787	0.3025
Medium SOC in Power shortage	0.2918	0.2775	0.3062
Low SOC in Power shortage	0.3010	0.2603	0.3417
High SOC in Power equilibrium	0.3457	0.3318	0.3595
Medium SOC in Power equilibrium	0.3244	0.3075	0.3412
Low SOC in Power equilibrium	0.3771	0.3557	0.3986

After that, considering the improvement in practical application, it is necessary to minimize the charging station improvement workload by combining the existing charging station construction base. Therefore, this paper analyzes the improvement benefits under different improvement workloads by combining the existing charging station construction base, and the results obtained are shown in Fig. 7.

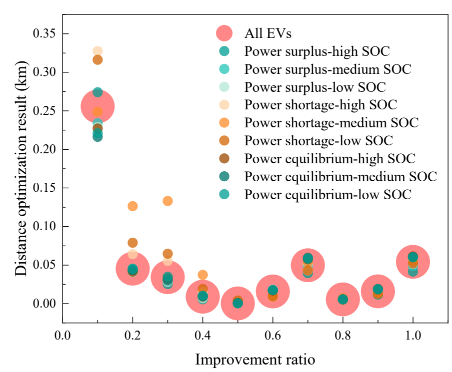


Fig. 7. Optimization effects of charging station improvements

It can be seen that when improving 10% of charging stations, the optimization effect growth rate reaches the maximum. When the improvement amount is greater than 10%, the effect is still improved, but the growth rate of improvement is smaller. Therefore, when improving the charging stations in Chicago, it can be prioritized to improve 10% of charging stations.

In summary, the charging stations planning method considering VPP aggregation proposed in this paper not only effectively reduces the overall average distance from EVs to charging stations but also considers the demand of different grid states and the distribution of EVs in different SOC states. It effectively improves the enthusiasm of EVs to participate in VPP flexible scheduling.

B. Scheduling Simulation

First, calculate the incentive electricity price. The 2018 EU version of BAIC Motor uses a ternary lithium battery with a capacity of 45kWh. It can be charged and discharged about 1,200 times, and the cost is about 3,184 US dollars. Set kp=1, $P_{t,n}^{dis}$ =30kWh. It can be obtained:

$$M_t = M_t^0 + 0.1474 \tag{54}$$

During this period, 71,176MWh of electric energy was generated in the PJM grid, and 71,297MWh of electric energy was required by the grid. Assuming that the VPP operator in the Chicago area needs to undertake the dispatch task of 5MWh, the total rechargeable energy of charging stations in this dispatch is P^e= 2MWh, η^{Sta} =0.7, ε^0=200kWh. Then P^n=1.4MWh, N^{All}=4. Use the set charging stations responsiveness statistical evaluation index RcI_n^t to sort the planned charging stations, set η^{DAl} =0.01, the sorting situation is as shown in Table IV.

TABLE IV
CHARGING STATIONS SORT RESULTS

Sorting result	High SOC average distance (km)	Medium SOC average distance (km)	low SOC average distance (km)	Number of EVs with high SOC	Number of EVs with medium SOC	Number of EVs with low SOC	RcI	Charging station number
1	0.6738	0.7587	0.5689	385	4	4	298.6425	14
2	0.2667	-	0.1779	103	-	1	93.1178	114
3	0.4542	0.6383	0.6383	129	4	7	80.5819	116
4	0.1080	0.0632	0.0632	84	2	1	74.2038	118
5	0.5031	-	0.3863	43	-	2	29.3204	171
6	0.1977	-	-	26	-	-	6.2296	154

It can be seen that the four charging stations, 14, 114, 116, and 118, were selected as the scheduling objects of this VPP. Since this time period is grid power shortage and the content of high SOC in this sample is much higher than that of low and medium SOC, the ranking is therefore almost consistent with the number of high SOCs around charging stations. But among the second and third charging stations, the number of high SOCs around the second charging station No. 114 is only 103, which is lower than the number of high SOCs around the third charging station No. 116, which is 129. This is because RcI_n^t not only considers the number of EVs, but also considers the distances from EVs to charging stations in different SOC states. The average distance from EVs to charging stations in the high SOC state around the second charging station is 0.2667 km, which is significantly lower than the average distance of 0.4542 from the third charging station. Therefore, distance information becomes the dominant factor at this time, making charging station No. 114 rank ahead of charging station No. 116. At the same time, there are 7 EVs in the low SOC state around No. 116 charging stations, while No. 114 has only 1 EV in the low SOC state, which is one of the reasons why No. 114 charging stations are ranked higher than No. 116 charging stations.

Table V demonstrates that the proposed charging occupancy control strategy effectively limits the regular charging utilization rate of dispatch-participating stations to below 30% during the dispatch period, in accordance with the dispatch task requirement of η^{Sta} =0.7. The utilization rates of high-load stations, such as Station 14 and Station 116, are significantly reduced, preventing the occupation of charging piles from affecting the completion of dispatch tasks. Meanwhile, a slight increase in the utilization of nearby stations indicates that the excess demand has been successfully redirected, thereby validating the effectiveness of the proposed charging guidance strategy in suppressing unnecessary pile occupation.

TABLE V
CHARGING STATION OCCUPANCY COMPARISON

Station ID	Occupancy (Regular)	Occupancy (Dispatch)	Avg. Occupancy of Nearby Stations (Regular)	Avg. Occupancy of Nearby Stations (Dispatch)
14	0.65	0.30	0.47	0.55
114	0.14	0.12	0.33	0.35
116	0.45	0.28	0.38	0.42
118	0.38	0.25	0.36	0.39

Note: Occupancy = $P_n^{occ} \ / \ \left(1-\eta^{Sta}\right)P_n^e$. Nearby station values refer to the average occupancy of non-dispatchable stations within 1 km.

After that we use the scheduling method proposed in this paper to compare the scheduling results of charging station planning using [17], [46]and [47] respectively. The charging stations are sorted according to RcI and the results of selecting the top four charging stations with the highest RcI are shown in Table VI. It can be seen that in the scheduling results of the proposed method in this paper, the RcI of each charging station is generally higher than the RcI of several other methods. And the sum of the top four high RcI of the proposed method in this paper is 546.546 which significantly higher than that of the other three methods. Therefore, the proposed method in this paper can increase the schedulable performance of the charging station (increase the number of dispatchable EVs near the charging station and decrease the distance of EVs participating in scheduling), which verifies the advantages of the proposed method in this paper.

TABLE VI
COMPARATIVE ANALYSIS OF SCHEDULING RESULT

	Scheduling the result of this paper	Scheduling the result of [17]	Scheduling the result of [45]	Scheduling the result of [46]
1st	298.64	183.12	146.52	216.76
2nd	93.11	85.19	115.52	117.76
3rd	80.58	68.99	66.72	51
4th	74.20	67.62	61.02	48.76
Sum	546.54	404.93	389.79	434.28

To sum up, using RcI_n^t to conduct real-time statistical analysis of EVs around charging stations can reasonably describe the charging capacity of each charging station, so that the EVs can be aggregated into VPP in sequence, to respond to electricity market demand.

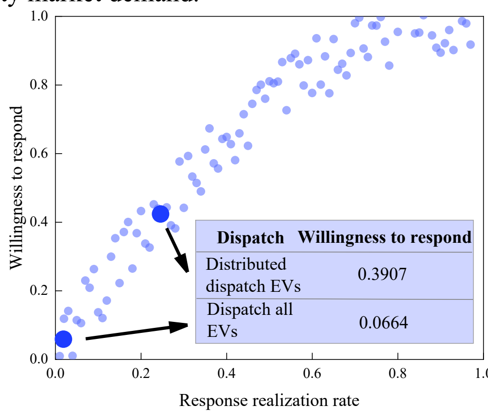


Fig.8. EV response willingness under different dispatch

Assuming that all online EVs are notified by regular scheduling, we calculate the average online EVs of 162,945 per hour, and obtain the response realization rate of the proposed distributed scheduling method and the regular method in this scheduling as shown in Fig. 8. The response willingness of the proposed method is 0.3907, and the response willingness of scheduling all EVs is 0.0664, which shows that the proposed method can reduce the number of notified EVs and thus improve the response willingness of EVs.

To validate the effectiveness of the flexible pricing mechanism and further demonstrate the advantages of the proposed method in enhancing EV user response willingness, a comparative analysis was conducted between two dispatch strategies. The scheme employing the optimized charging station layout and distributed dispatch strategy proposed in this study is referred to as Proposed, while the scheme using the original station layout and broadcasting dispatch signals to all online EVs is referred to as Baseline. Assuming an initial incentive price of 1.1474 USD/kWh, if the predicted response energy fails to meet the dispatch demand, the incentive price is increased in successive rounds by a fixed increment of 0.20 USD/kWh until the expected target is achieved.

TABLE VI

COMPARISON OF SCHEDULING PROCESSES

Iteration	Price (USD / kWh)	Proposed Response (MWh)	Baseline Response (MWh)
0	1.1474	4.83	4.68
1	1.3474	5.04	4.92
2	1.5474	5.34	5.13
3	1.7474	—	5.37

The experimental results indicate that although the initial dispatch responses under both strategies failed to meet the dispatch requirements, the flexible price adjustment mechanism enabled both strategies to eventually achieve the dispatch target of 5.2 MWh through multiple rounds of price adjustments, thereby validating the effectiveness of the proposed mechanism. Specifically, the final dispatch price under the Proposed strategy was 1.5474 USD/kWh, while the Baseline strategy required a higher final dispatch price of 1.7474 USD/kWh. This demonstrates that under the same dispatch demand, the proposed strategy incurs lower dispatch costs. It also shows that the proposed strategy can enhance user participation willingness, lower the price threshold at which EVs are willing to respond, and thus improve the economic efficiency of the dispatch process.

VI. Conclusion

This paper studies the scheduling problem of EVs. A dispatching model that takes into account the willingness to respond is proposed. The key findings are as follows.

- A three-step weighted k-mean clustering algorithm is proposed by considering the relationship between grid demand and EVs of different SOC states. The algorithm reduces the consumed travel distance needed for EVs to respond to grid dispatch, reducing the average distance from EVs to charging stations by 0.188km in the city of Chicago.
- A distributed scheduling strategy centred on charging stations is proposed. The approach analyzes the state of EVs during scheduling, realizes a sequential scheduling strategy that considers multiple factors, and increases the willingness of EVs to participate from 0.0664 to 0.3907 by improving their response realization rate.

This study addresses key challenges in EV scheduling, including range anxiety, battery aging, and low response rates, by proposing methods for improved charging station siting, incentive pricing, and distributed scheduling. The robustness of the scheduling mechanism is critical for accurately utilizing electric vehicle charging scenarios, as communication failures or unexpected behaviors can impact the effectiveness of the proposed method. The introduction of scheduling margin ensures system reliability even under uncertain conditions, highlighting the importance of robust scheduling in practical deployment.

Future research could further explore the impact of social-psychological factors and government policies on electric vehicle scheduling decisions, incorporating behavioral decision theories and analyzing the effects of policy frameworks. Additionally, more in-depth studies on robust scheduling will be conducted, particularly focusing on strategies to mitigate large-scale communication failures or other extreme scenarios.

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